PROBLEM - I

For the given beam with an applied bending moment (M_0)

FIND

(i) σ_{xx}

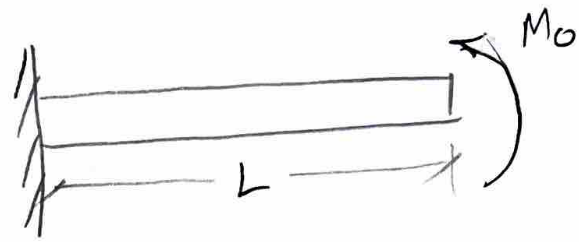
(ii) $\sigma'_{xx})_{max}$: the maximum normal stress

(iii) $\sigma'_{xy})_{max}$: the maximum shear stress

(iv) deflection curve

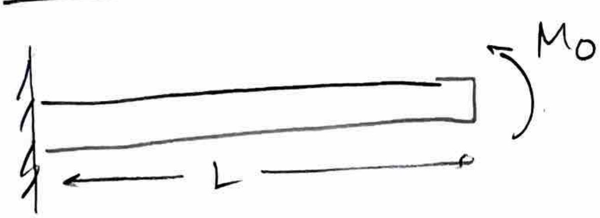
<u>Note</u>:

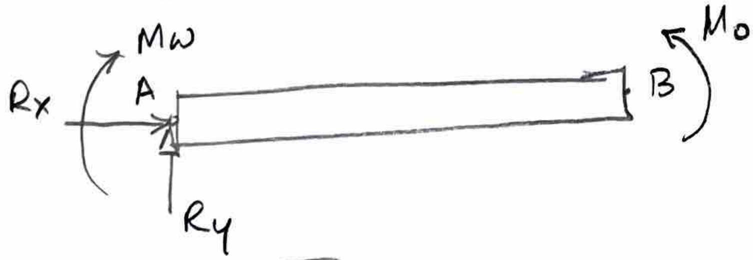
The given beam is a cantilever beam with a rectangular cross-section.



SOLUTION:

(WHOLE)
<u>FREE BODY DIAGRAM</u>





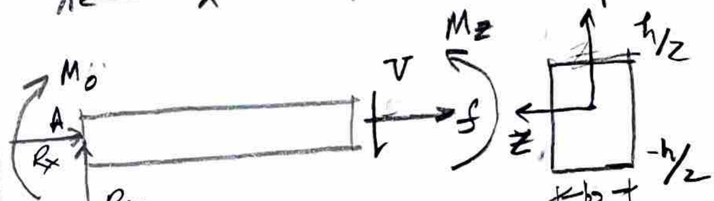
$\boxed{\Sigma F_x = 0} \Rightarrow R_x = 0$

$\boxed{\Sigma F_y = 0} \Rightarrow R_y = 0$

$\boxed{\Sigma M_z)_B = 0} \Rightarrow M_0 - M_W = 0$

$\Rightarrow M_0 = M_W$

(PARTIAL)
<u>FREE BODY DIAGRAM</u>



$\boxed{\Sigma F_x = 0} \Rightarrow \cancel{R_x}^{0} + f = 0$

$\Rightarrow f = 0$

$\boxed{\Sigma F_y = 0} \Rightarrow \cancel{R_y}^{0} - V(x) = 0$

$\Rightarrow V(x) = 0$

$\boxed{\Sigma M_z)_A = 0} \Rightarrow -M_0 + M_z(x) - \cancel{V(x)}^{0} = 0$

$\Rightarrow M_z(x) = M_0$

Flexure formula

(i) $\boxed{\sigma_{xx} = -\frac{M_z(x)\, y}{I_{zz}}}$ $\Rightarrow$ $\boxed{\sigma_{xx} = -\frac{M_0 y}{I_{zz}}}$

since we know $M_z(x) = M_0 \quad \forall\, x \in [0, L]$

$y \in [-\frac{h}{2}, \frac{h}{2}]$

$z \in [-\frac{b}{2}, \frac{b}{2}]$

$$I_{zz} = \int y^2 dA$$

$$\Rightarrow I_{zz} = \int_{-b/2}^{b/2} \int_{-h/2}^{h/2} y^2 \, dy\, dz$$

$$= \int_{-b/2}^{b/2} \frac{y^3}{3}\Big|_{-h/2}^{h/2} dz = \int_{-b/2}^{b/2} \frac{\cancel{2}}{3}\frac{h^3}{\cancel{8}4} dz$$

$$= \frac{1}{12} h^3 z \Big|_{-b/2}^{b/2} = \frac{1}{12} h^3 \cancel{2} \frac{b}{\cancel{2}}$$

$$\Rightarrow \boxed{I_{zz} = \frac{h^3 b}{12}}$$

$$\Rightarrow \sigma_{xx} = -\frac{M_0 y}{h^3 b/12} \Rightarrow \boxed{\sigma_{xx} = -\frac{12 M_0 y}{h^3 b}}$$

✓

(ii) Largest compressive & tensile loads relative to (x, y, z) occur at $y = \pm \frac{h}{2}$

$$\sigma_{xx}\left(y = \pm \frac{h}{2}\right) = \mp \frac{\overset{6}{\cancel{12}} M_0}{h^{\cancel{3}2} b}\left(\frac{\cancel{h}}{\cancel{2}}\right)$$

$$\Rightarrow \boxed{\sigma_{xx}\left(y = \pm\frac{h}{2}\right) = \mp \frac{6 M_0}{h^2 b}}$$

Min
Max., normal stress ⇝ Principal stresses

$$\sigma'_{xx}\Big)_{\substack{max\\min}} = \frac{\sigma_{xx}+\sigma_{yy}}{2} \pm \sqrt{\left(\frac{\sigma_{xx}-\sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2}$$

considering 2D state of stress σ_{xx}, σ_{yy} & $\sigma_{xy}=\sigma_{yz}$

where $\sigma_{yy} = \sigma_{xy} = 0$

$$\Rightarrow \sigma'_{xx}\Big)_{max} = \frac{\sigma_{xx}}{2} + \sqrt{\frac{\sigma_{xx}^2}{4}}$$

$$= \frac{\sigma_{xx}}{2} + \frac{\sigma_{xy}}{2} \Rightarrow \boxed{\sigma'_{xx}\Big)_{max} = \frac{6M_o}{h^2 b}}$$

$$\Rightarrow \sigma'_{xx}\Big)_{min} = \frac{\sigma_{xy}}{2} - \sqrt{\frac{\sigma_{xx}^2}{4}} = \frac{\sigma_{xx}}{2} - \frac{\sigma_{xy}}{2} = 0$$

$$\Rightarrow \boxed{\sigma'_{xx}\Big)_{min} = 0}$$

Also $\alpha_p = 0$ i.e., x, y in principal directions

(iii) **Max., Min sheer stress:**

$$\sigma'_{xy}\Big)_{\substack{max\\min}} = \pm\sqrt{\left(\frac{\sigma_{xx}-\sigma_{yy}}{2}\right)^2 + \sigma_{xy}^2}$$

Again
$\sigma_{yy} = 0$
$\sigma_{xy} = 0$

$$= \pm \frac{\sigma_{xx}}{2}$$

$$\Rightarrow \sigma'_{xy}\Big)_{\substack{max\\min}} = \pm \frac{\overset{3}{\cancel{6}} M_o}{\cancel{2} h^2 b} \Rightarrow \boxed{\sigma'_{xy}\Big)_{\substack{max\\min}} = \pm \frac{3M_o}{h^2 b}}$$

Also $\alpha_s = 45^\circ$

(iv) Deflection curve :

Let $v(x)$ be the deflection curve or shape of the neutral axis in deformed configuration

Governing equation : $\boxed{EI_{zz}\dfrac{d^2v}{dx^2} = M_z(x)}$

We know already $M_z(x) = M_0$

$$\therefore \boxed{EI_{zz}\frac{d^2v}{dx^2} = M_0}$$

$$\Rightarrow \frac{d^2v}{dx^2} = \frac{M_0}{EI_{zz}}$$ Integrating

$$\int \frac{d}{dx}\left(\frac{dv}{dx}\right) dx = \frac{M_0}{EI_{zz}} \int dx$$

$$\Rightarrow \frac{dv}{dx} = \frac{M_0}{EI_{zz}} x + C_1$$

Integrating

$$\int \frac{d\,v(x)}{dx} dx = \int \left(\frac{M_0}{EI_{zz}} x + C_1\right) dx$$

$$\Rightarrow \boxed{v(x) = \frac{M_0 x^2}{2EI_{zz}} + C_1 x + C_2}$$

$$\boxed{v(x) = \frac{M_0 x^2}{2EI_{zz}} + C_1 x + C_2}$$

Quadratic eq.

↳ we need two boundary conditions

BC (I) $\boxed{\frac{dv}{dx} = 0 \text{ at } x = 0}$

$$\Rightarrow \frac{dv}{dx} = \frac{\cancel{2} M_0 x}{\cancel{2} EI_{zz}} + C_1 = 0 \quad \text{at } x = 0$$

$$\Rightarrow \frac{M_0 (0)}{EI_{zz}} + C_1 = 0 \Rightarrow \boxed{C_1 = 0}$$

BC (II)

$$v(x) = 0 \quad \text{at } x = 0$$

$$\Rightarrow v(x) = \frac{M_0 x^2}{2EI_{zz}} + C_2 = 0 \quad \text{at } x = 0$$

$$\Rightarrow \frac{M_0 (0)}{2EI_{zz}} + C_2 = 0$$

$$\Rightarrow \boxed{C_2 = 0}$$

We know $I_{zz} = \frac{bh^3}{12}$ for rectangular cross-section

$$\boxed{v(x) = \frac{6 M_0 x^2}{E b h^3}}$$

↳ Deflection curve

Deflection at $x = L$: $\delta = v(x = L)$
(δ)

$$\Rightarrow \boxed{\delta = \frac{6 M_0 L^2}{E b h^3}}$$